

Literature Benchmark Brief: Karniadakis_2021_NatRevPhys

Title: Physics-informed machine learning

Authors: G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, L. Yang (2021)

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Cluster: Cluster E: Missing Data & Modular Risk Layers

Core Theoretical Mechanism & Model Formulation

Tripartite inductive bias taxonomy: Observational bias, Inductive architectural bias, Learning bias

Empirical Scale & Experimental Dataset

Multiscale physical systems and engineering fluid dynamics

Relevance to IEEE TSTE & Sustainable Energy Systems

Highest authority for methodology classification in physics-informed AI

Identified Gaps & Operational Vulnerabilities

General review; does not formulate industrial cyber-physical SCADA telemetry latency

Role in Manuscript Defense & Reviewer #2 Proof

Classifies our wake graph as Inductive bias and alignment loss as Learning bias